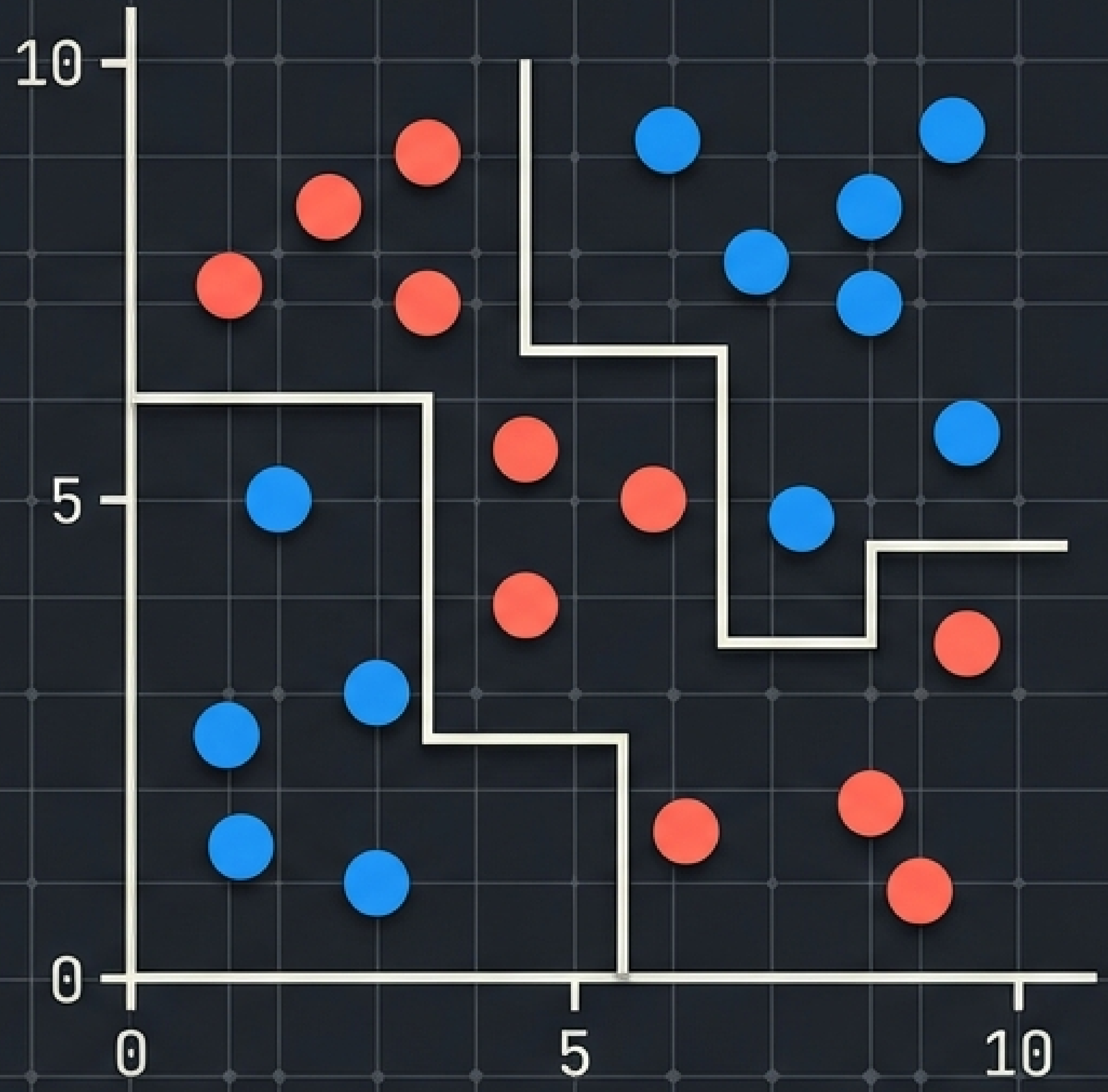
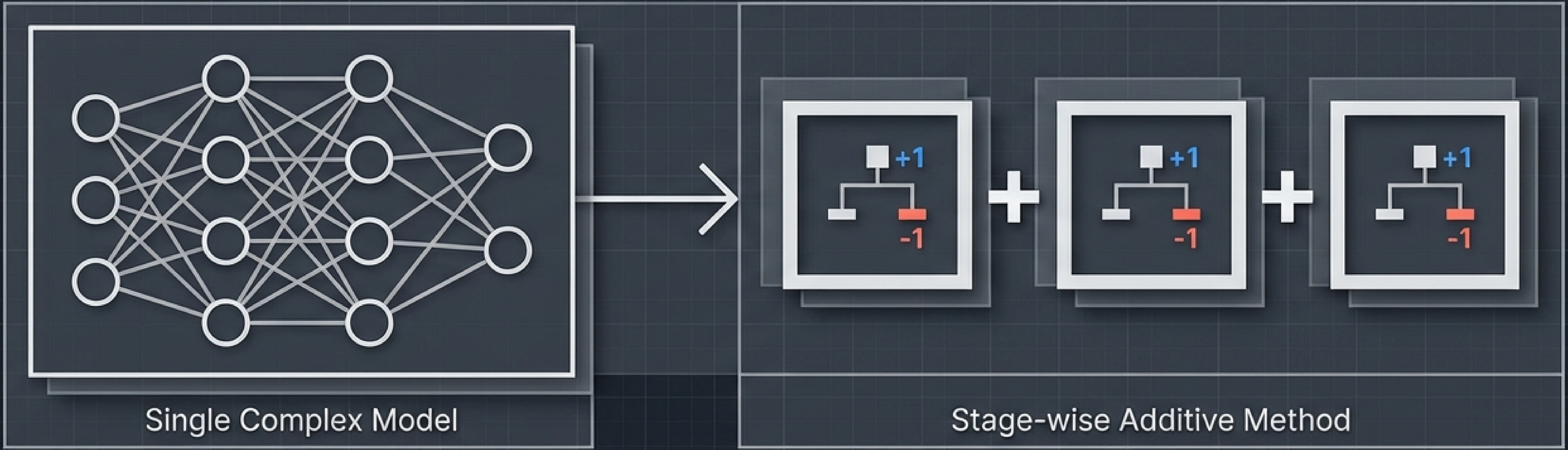


The Geometric Blueprint of AdaBoost

Visualising the mathematics of sequential ensemble learning

A 25-year-old foundation for modern boosting



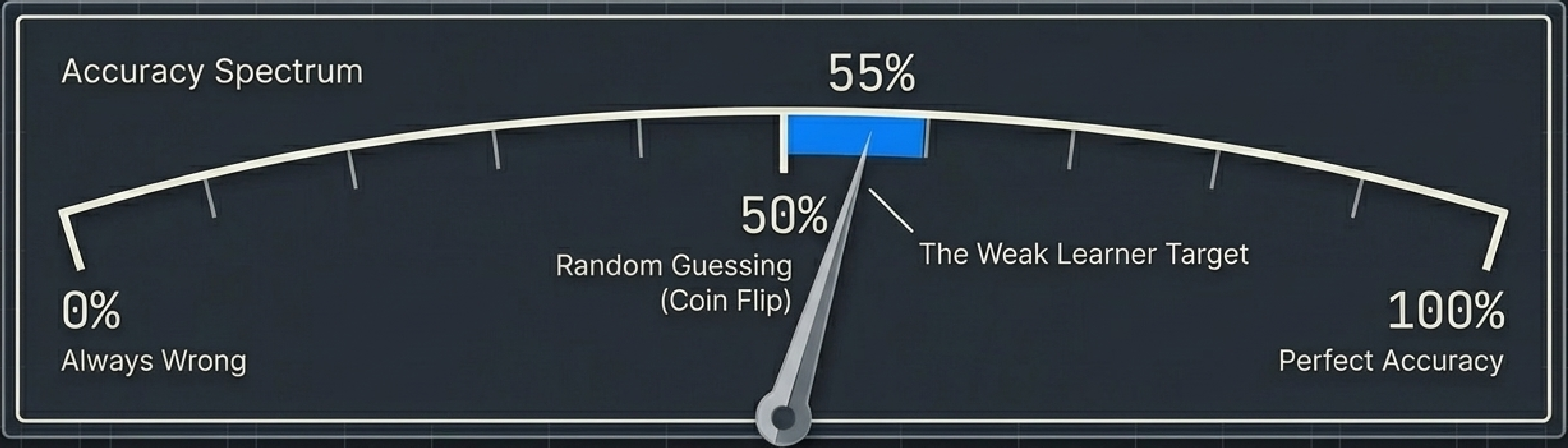
- AdaBoost (Adaptive Boosting) remains the conceptual gateway to all modern boosting algorithms.
- Instead of training one massive model, it sequentially adds simple models together.
- Each new model strictly focuses on the errors made by the previous one.

Shifting from binary labels to mathematical signs

Standard Classification		AdaBoost Format	
0	1	-1	+1
JetBrains Mono	JetBrains Mono	JetBrains Mono	JetBrains Mono
		Negative / Not Placed	Positive / Placed

- AdaBoost abandons traditional 0 and 1 labels.
- Positive outcomes are strictly mathematically coded as +1.
- Negative outcomes are strictly mathematically coded as -1.
- This symmetric geometry allows the final algorithm to use positive and negative numbers to tally mathematical votes.

The prerequisite of a Weak Learner

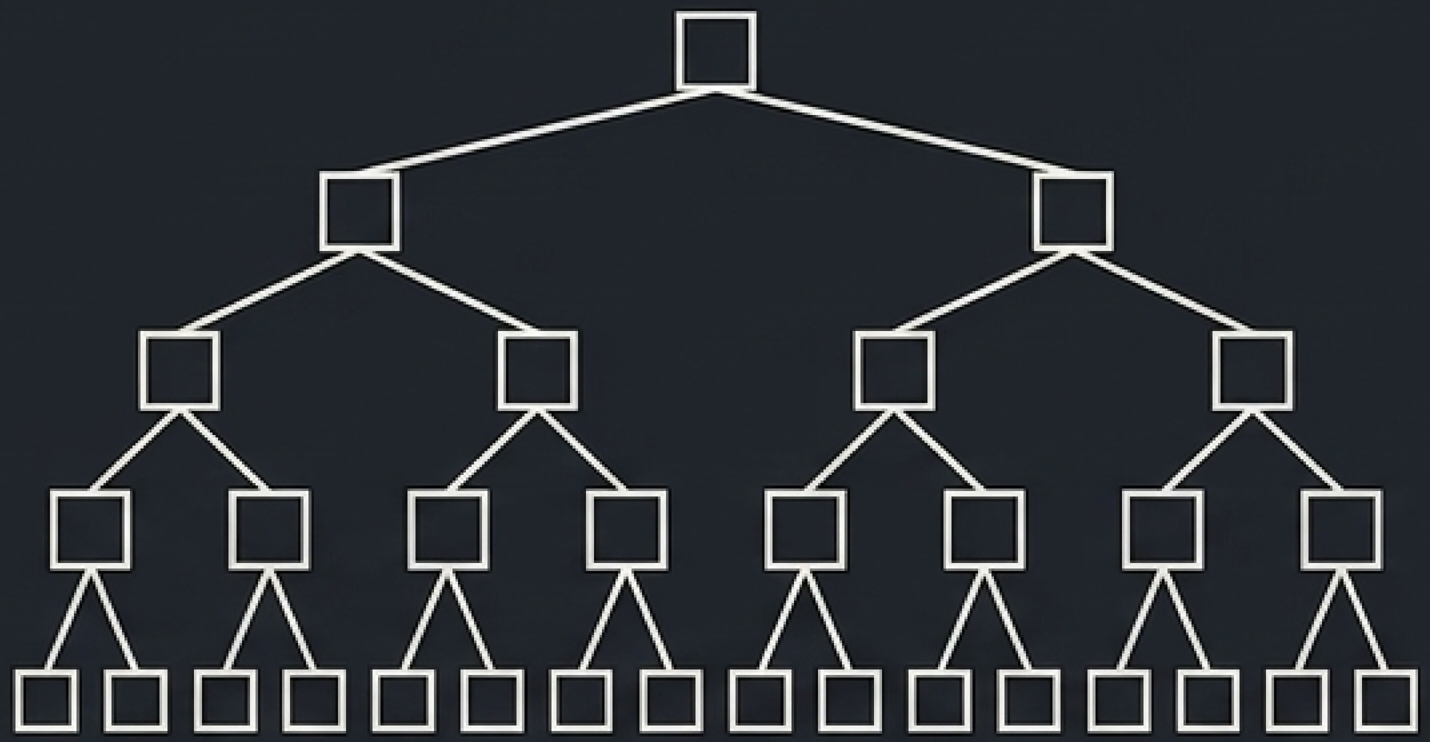
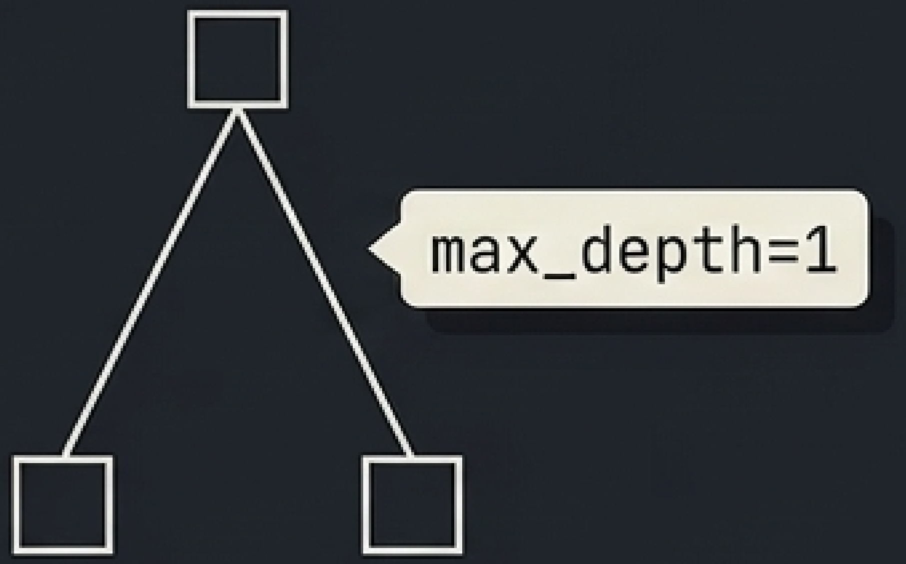


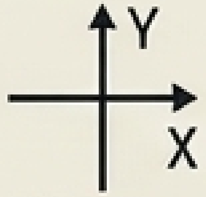
AdaBoost does not require complex base models.

It relies on Weak Learners: models whose accuracy is only slightly better than random guessing.

Individually, they are highly error-prone. Collectively, they become exceptionally strong.

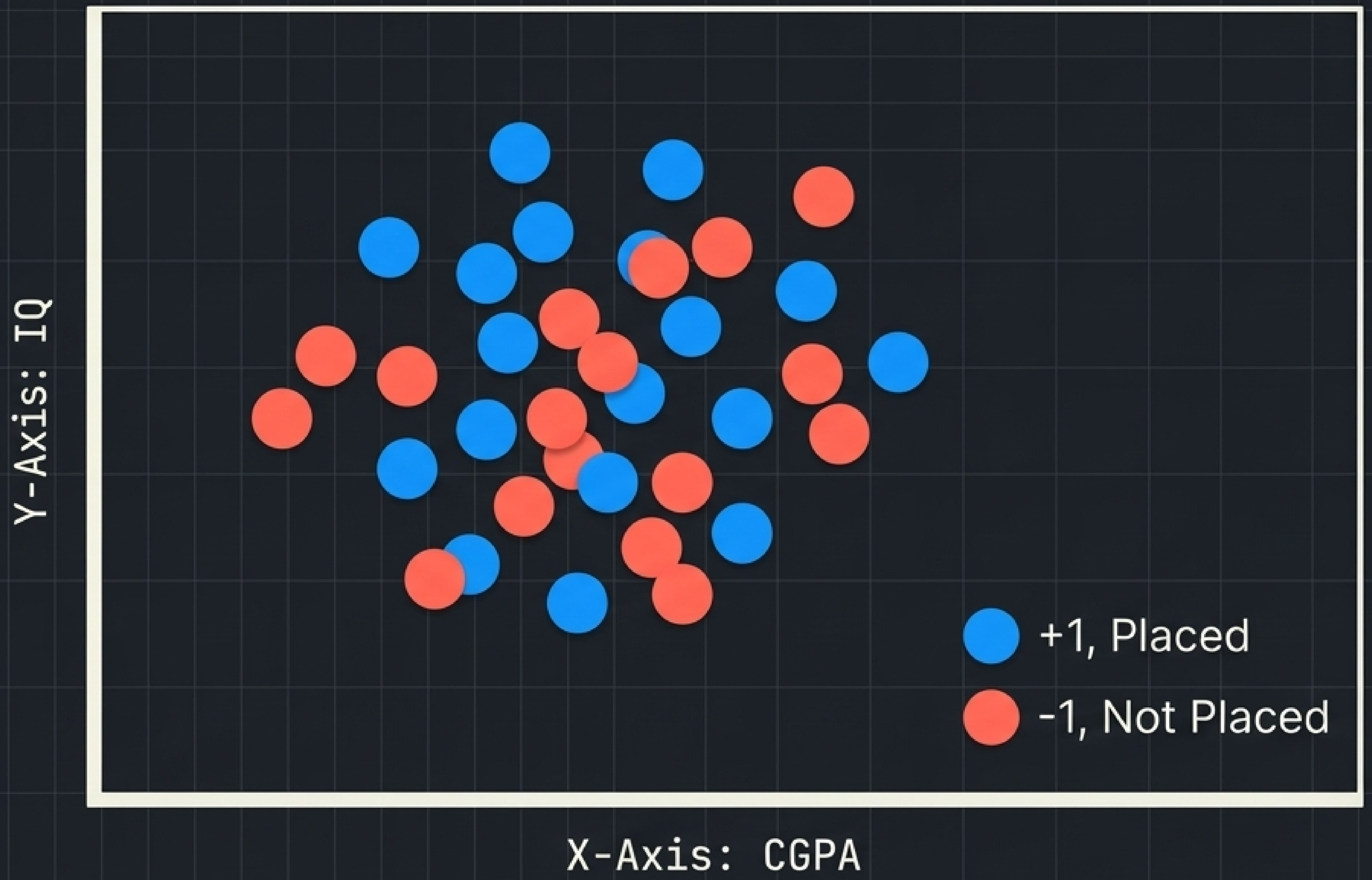
Constraining logic to a single Decision Stump

Standard Decision Tree	Decision Stump
	
Deep logic, multiple splits, high accuracy.	A single rule, one split, weak accuracy.



THE GEOMETRIC REALITY: On a 2D plane, a Decision Stump is simply a single straight line drawn entirely parallel to either the X or Y axis. It cuts the data exactly once.

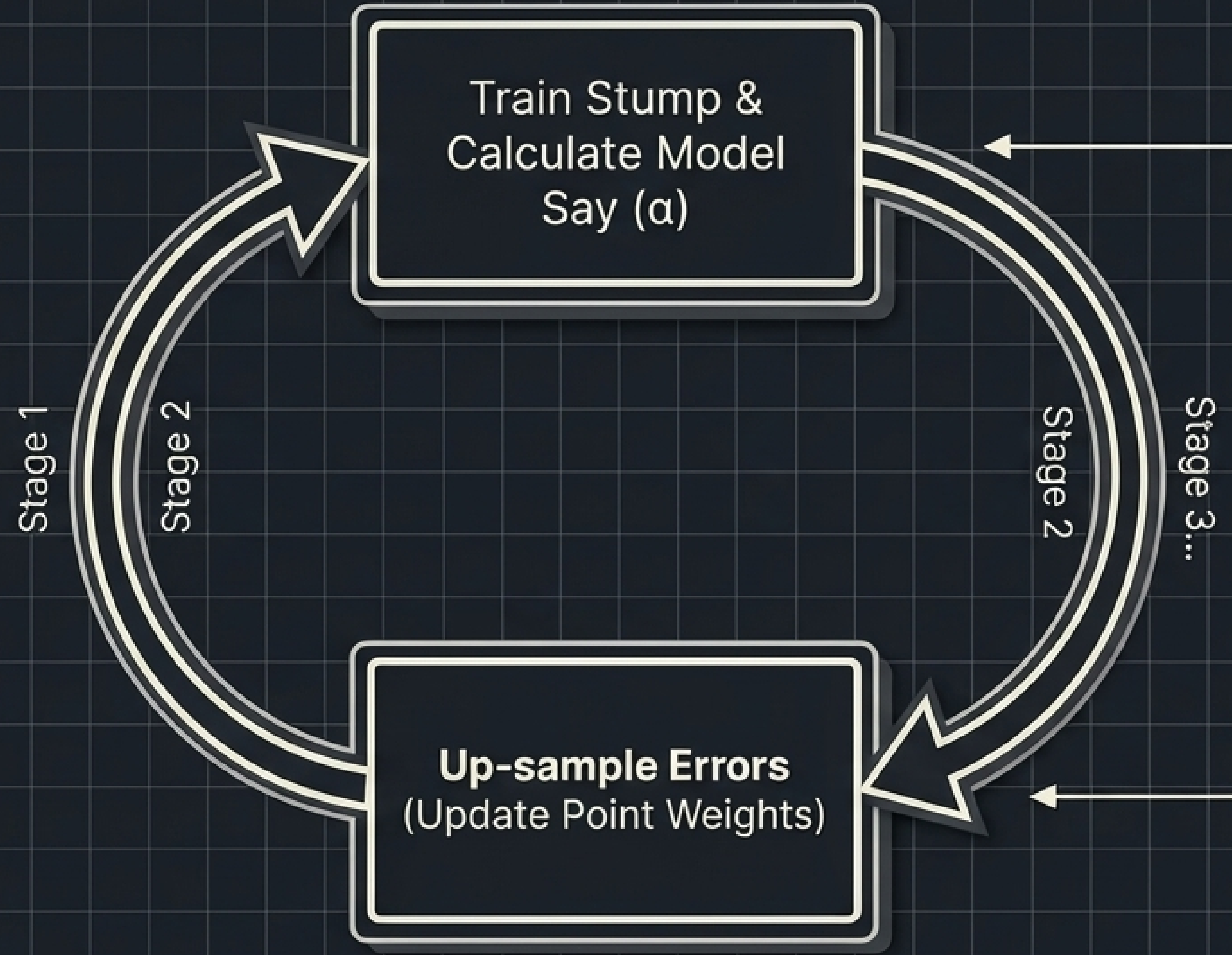
Plotting the raw classification space



The Goal:
Separate the Blue (+1) from the Coral (-1).

The Constraint:
We can only use straight, axis-parallel lines (Decision Stumps) to slice the canvas.

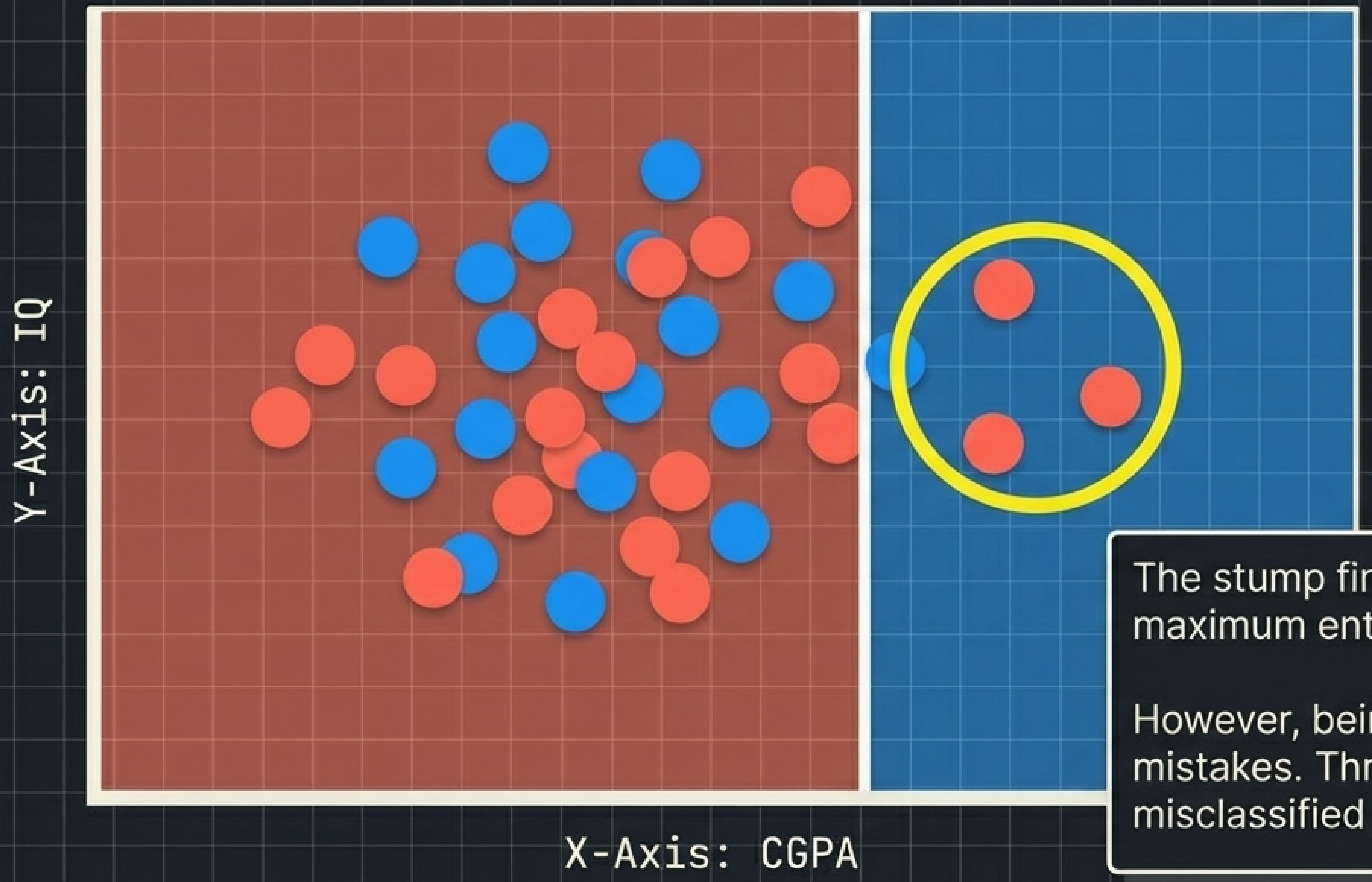
The two-step continuous feedback loop



Calculate α: The model's overall accuracy determines its voting weight in the final equation. High accuracy = high α.

Up-sample Errors: Misclassified data points physically increase in mathematical importance, forcing the next model to focus entirely on them.

Stage 1 makes the best single cut available

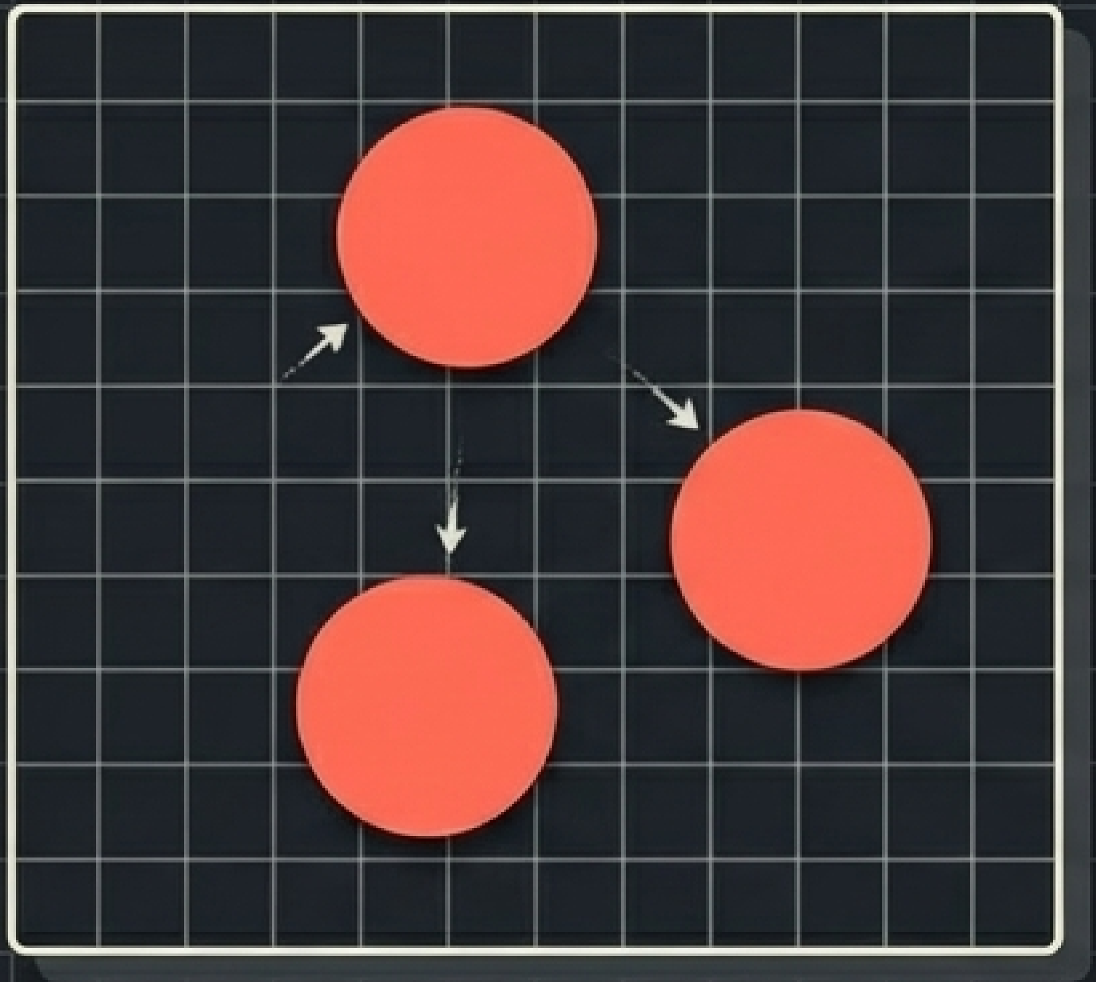


The stump finds the split that yields the maximum entropy reduction.

However, being a weak learner, it makes mistakes. Three negative points are misclassified as positive.

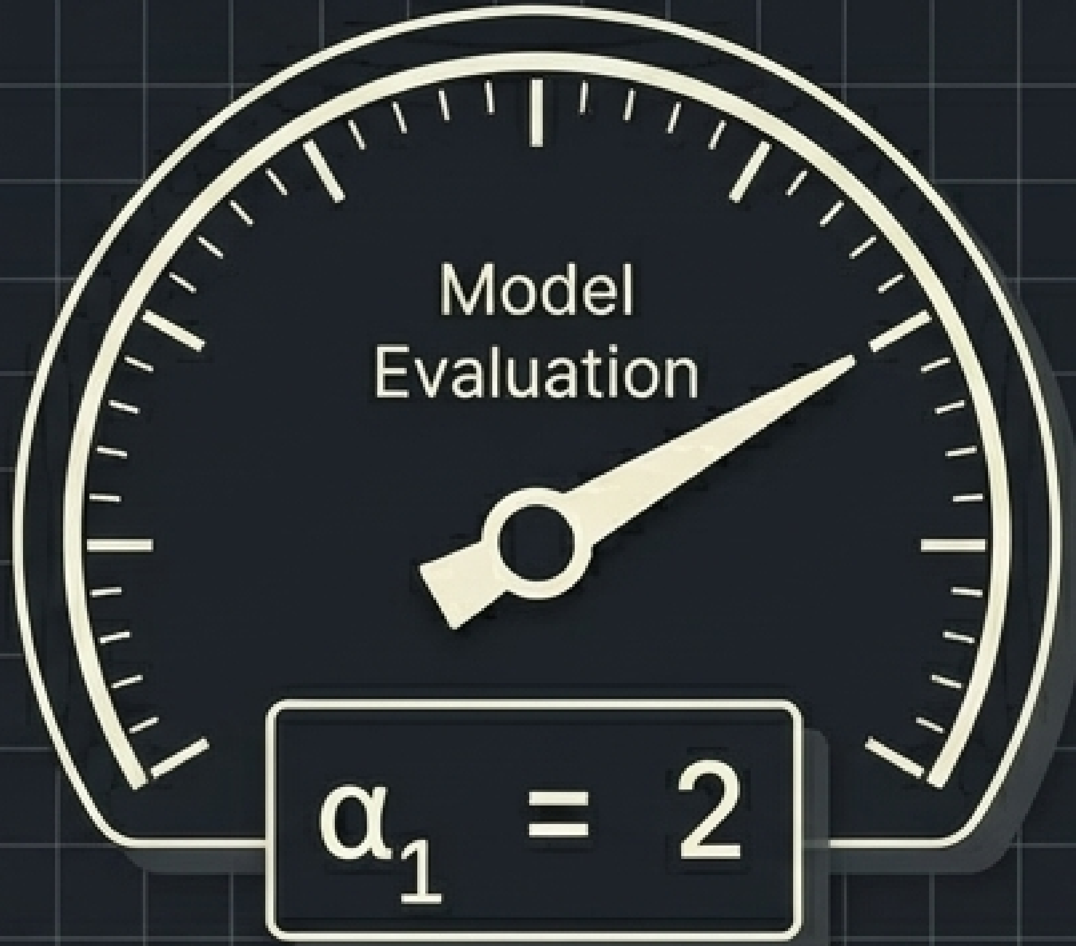
Penalising the mistakes to force adaptation

Up-sampling Misclassifications



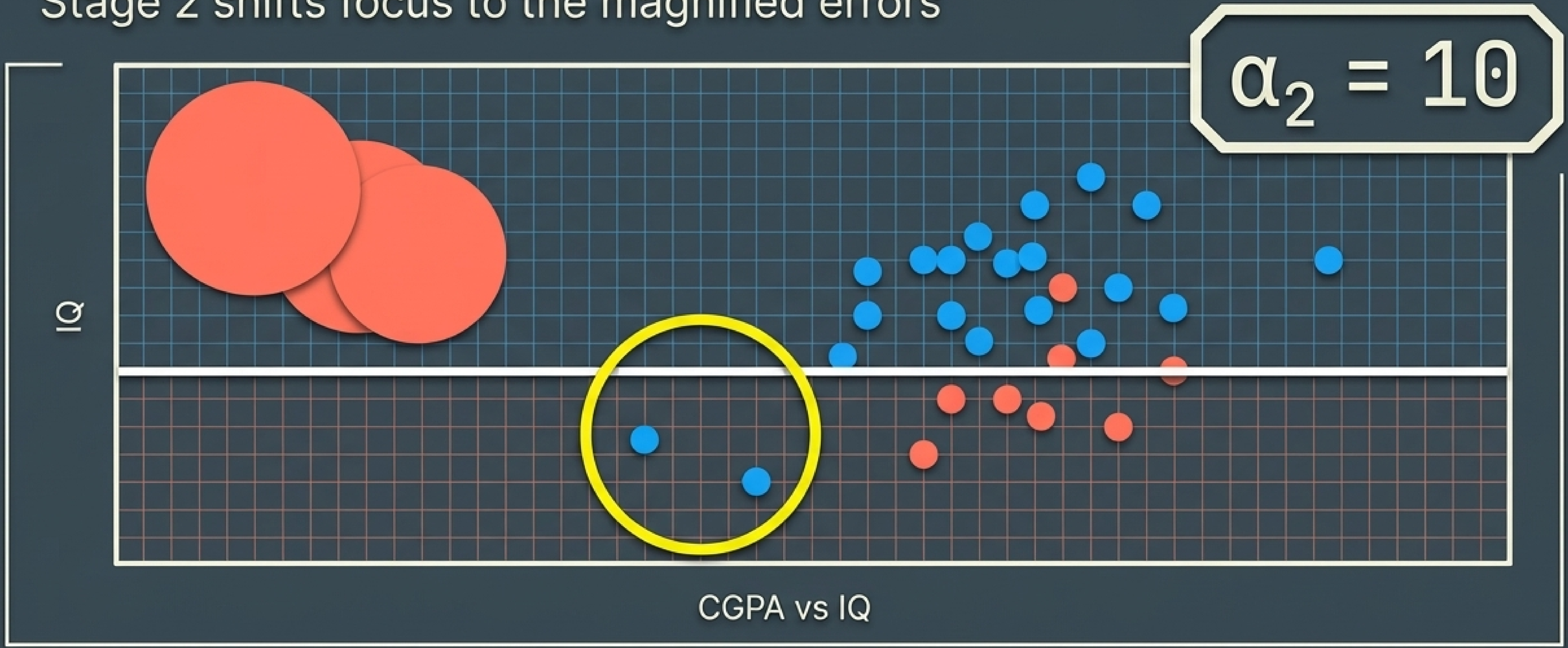
The misclassified points are up-sampled. Their data weight is increased, making them mathematically louder for the next round.

Model Evaluation



Stump 1 is evaluated. Based on its error rate, it is assigned a permanent voting weight: α_1.

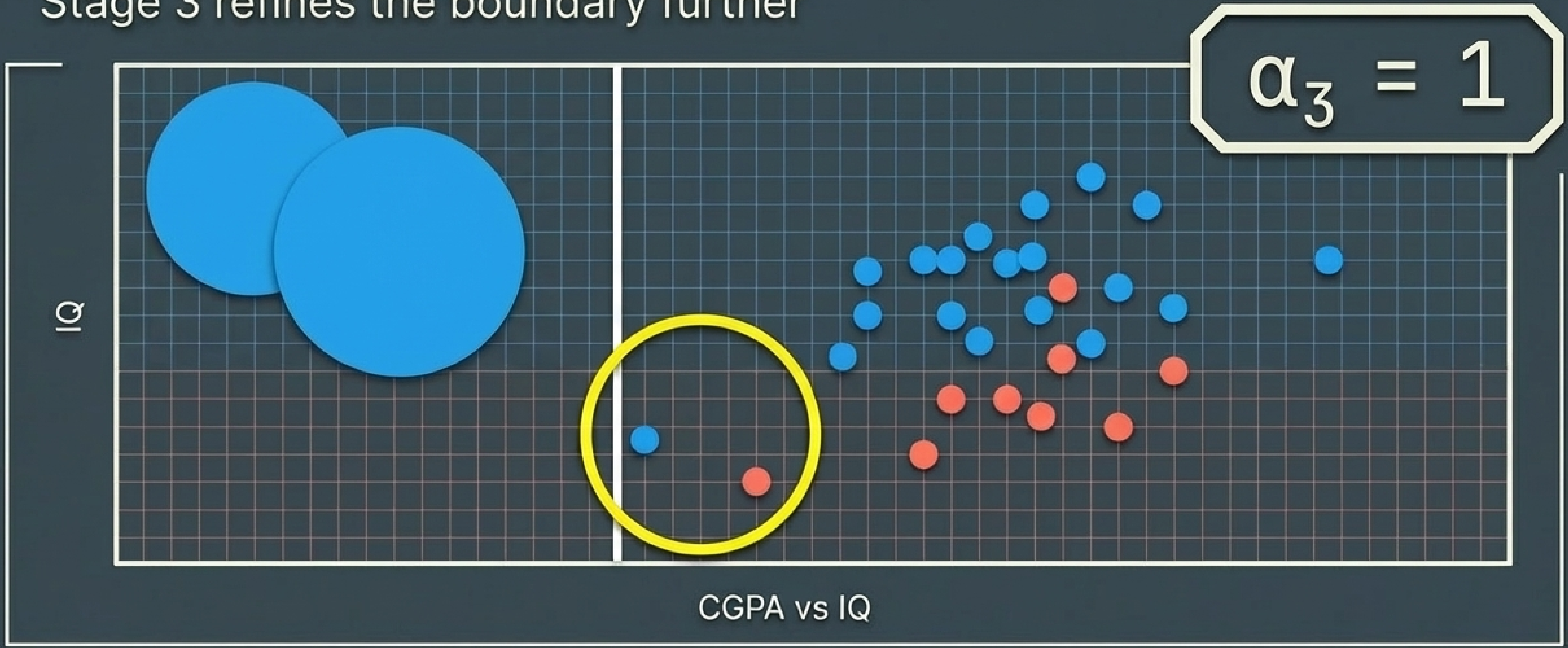
Stage 2 shifts focus to the magnified errors



Stump 2 completely ignores the overall dataset to ensure the three enlarged Coral points are captured correctly.

In doing so, it creates new, different mistakes. These new errors are up-sampled for Stage 3.

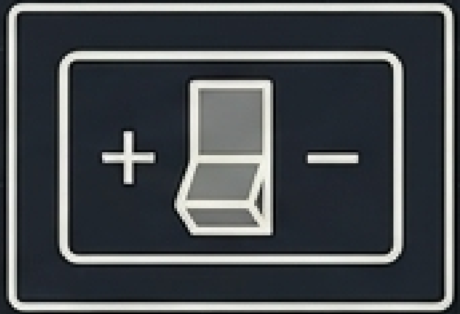
Stage 3 refines the boundary further



Stump 3 targets the new errors, generating its own unique slice of the data.

This sequential punishment of errors could continue for hundreds of stages, but we will stop at three to assemble the final model.

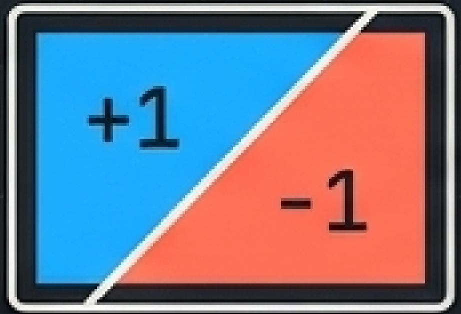
Assembling the additive equation



The Final Sign Tally

The Model's Earned Authority (Voting Power)

$$\text{Sign}(\alpha_1 M_1(x) + \alpha_2 M_2(x) + \alpha_3 M_3(x))$$



The Stump's Prediction (+1 or -1)

AdaBoost is simply a weighted sum. Each stump casts a base vote.	That vote is mathematically multiplied by the stump's earned authority (α).	If the final tally is positive, the prediction is +1. If negative, -1.

Tracing a single prediction through the system

New Profile - CGPA: 7.5 | IQ: 110

Stump 1 ($\alpha_1 = 2$) →	Votes -1 →	$2 \times (-1) = -2$
Stump 2 ($\alpha_2 = 10$) →	Votes +1 →	$10 \times (+1) = +10$
Stump 3 ($\alpha_3 = 1$) →	Votes -1 →	$1 \times (-1) = -1$

Final Tally Box

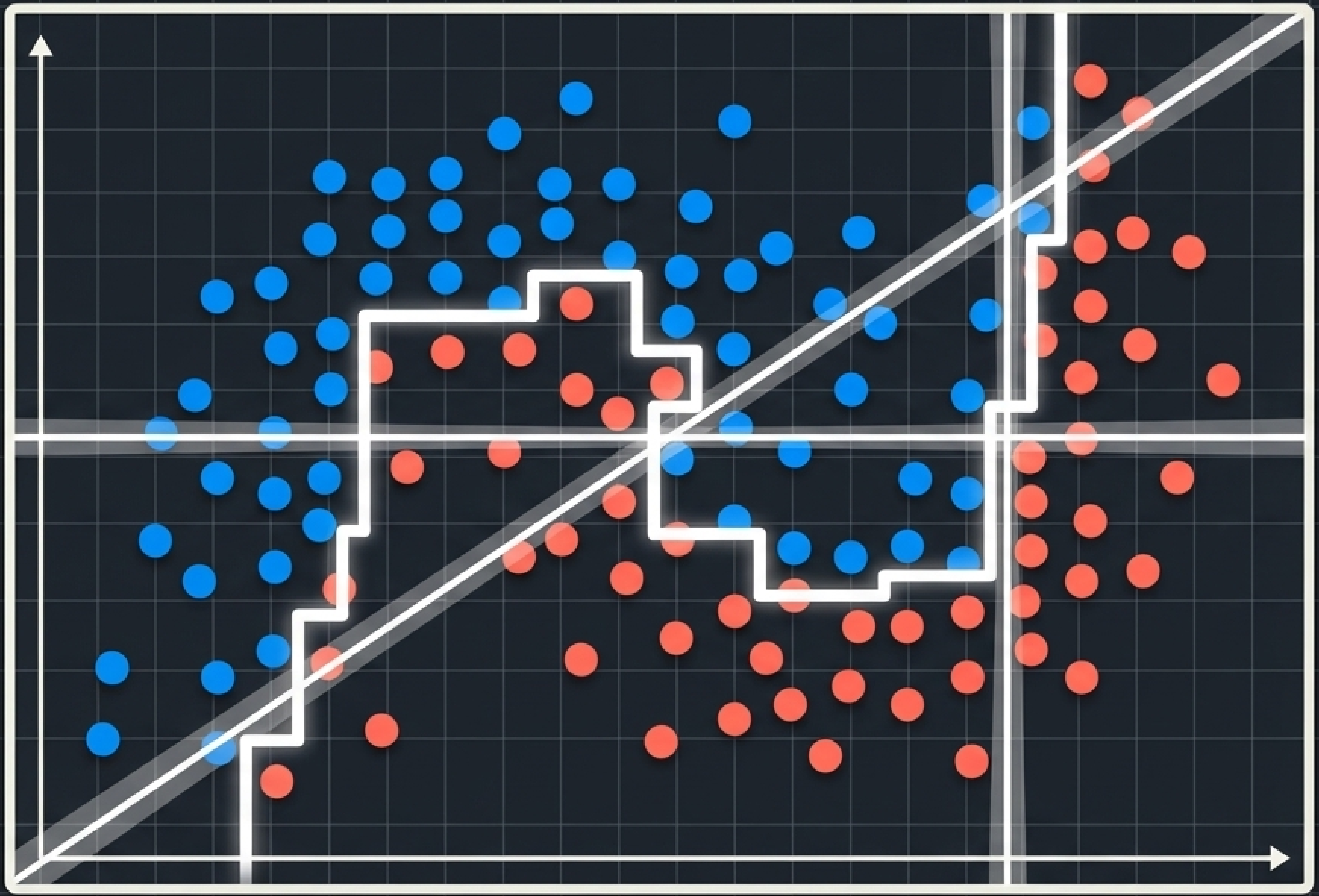
$$-2 + 10 - 1 = +7$$

Because **+7** is a positive number, the Sign function converts it to **+1**.

Final Prediction: Placed.

Notice how Stump 2's massive weight (α=10) overruled the negative votes of the other two models combined. Democracy, but weighted by competence.

The resulting geometric reality



No single straight line could classify this dataset.

But by superimposing multiple, weighted straight lines, AdaBoost artificially constructs a highly complex, non-linear decision boundary.

Complexity born from sequential simplicity

Weak Foundations

Starts with rigid, inaccurate stumps.

Guided by Mistakes

Driven entirely by up-sampling previous misclassifications.

Weighted Assembly

Predictions are settled by an authority-weighted vote.

This elegant combination of simple geometry and dynamic weighting is why AdaBoost remains a critical architecture in the history of machine learning.